Run report: `f4_fp` before and after #1717 in the Jacobian-cover index calculus, run-20261018T1548Z

Observations only. Interpretation: `../../REPORT.pdf`.

What ran

- **Runner:** `research/f4_fp_ic_effect_20261018/run.sh <jv_cover_main> <jv_cover_new> <this dir>`.
- **Arms:** release builds of `examples/jv_cover.rs` with `src/cryptanalysis/f4_fp.rs` as on `main` before #1717 (`4b6e09b78`, "before") and as merged in #1717 (branch `f4-fp-msolve-gap` at `3f5b08ff7`, "after"). SHA-256 in `raw/versions.txt`. Nothing else differs.
- **Pipeline:** Nagao decompositions on a genus-2 cover of $E(\mathbb{F}_{p^6})$; every decomposition test is one `f4_fp` solve (degree bound 24, budget 300 s).
- **Experiments** (same seeds in both arms):
  - `--exp ccov --sizes 53,101,251 --seeds 1 --residuals 100 --constructed 20` measures the cost of one decomposition test, C_{cov}: 100 random residuals plus 20 planted decomposable ones per prime, checked against a meet-in-the-middle oracle where the oracle is cheap.
  - `--exp dlp --sizes 53,101 --seeds 2 --rho-runs 4 --check-every 64` runs the DLP end to end: relations, linear algebra and verified logarithm, against 4 Pollard-rho runs on the same curve.
- **Order:** ccov before, ccov after, then dlp before, dlp after. The counted costs are deterministic; wall times are listed but not compared.
- **Host:** Claude Code cloud container, `Intel Xeon @ 2.80GHz`, 4 vCPU, shared, isolation L0.
- **Requested / completed:** 4 of 4 invocations completed, exit 0.

One decomposition test (ccov, means over 120 tests per prime)

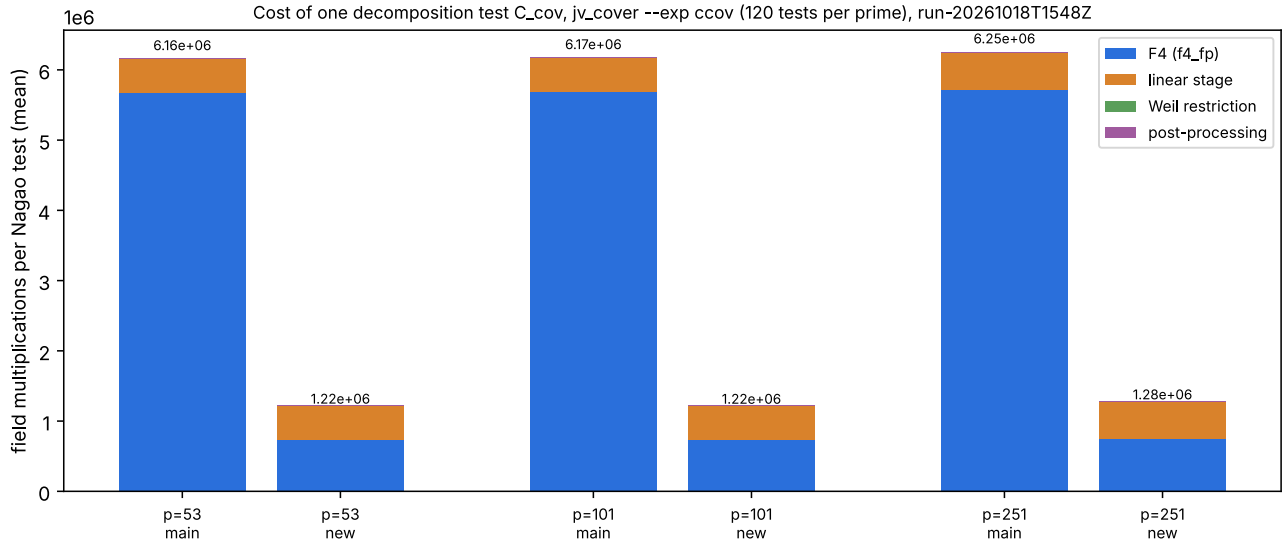


Figure 1: C_{cov} split by stage, run-20261018T1548Z.

p	planted found	C_{cov} before	C_{cov} after	F4 mults before ⟶ after	F4 degree	rows
53	20/20, 20/20	6.158e6	1.224e6	5.671e6 ⟶ 7.380e5	10.0 ⟶ 8.0	769 ⟶ 235
101	20/20, 20/20	6.172e6	1.223e6	5.686e6 ⟶ 7.369e5	10.0 ⟶ 8.0	767 ⟶ 231
251	20/20, 20/20	6.253e6	1.284e6	5.713e6 ⟶ 7.447e5	9.9 ⟶ 8.0	767 ⟶ 231

The linear stage is identical in both arms (4.82e5, 4.82e5, 5.35e5 mults). At p = 53 the oracle checked 120 tests per arm with 0 mismatches.

End to end (`dlp`)

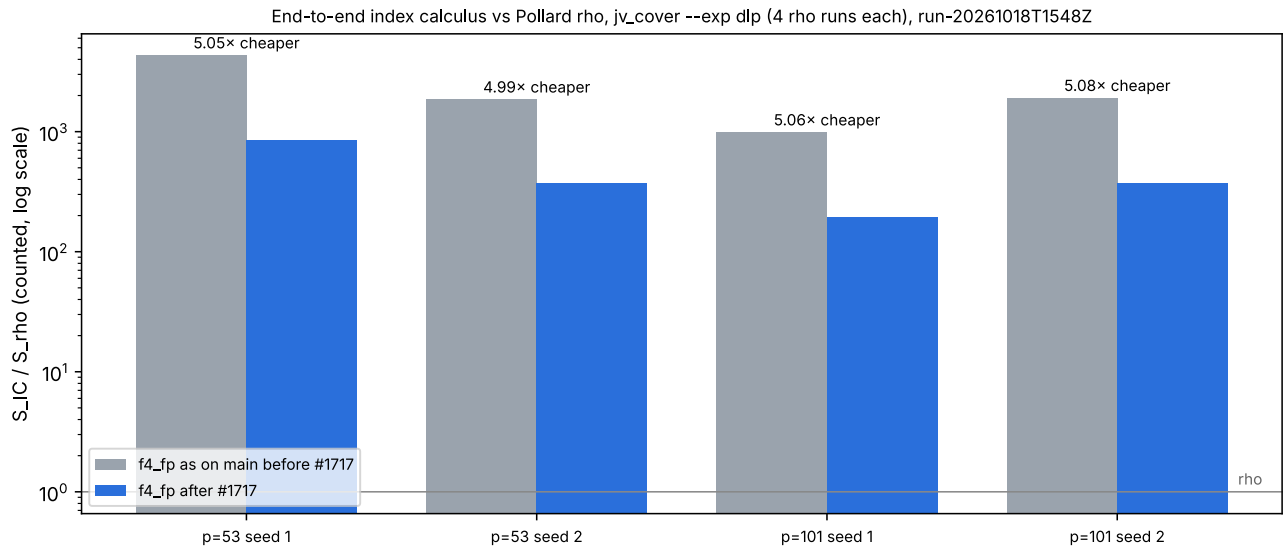


Figure 2: $S_{\mathrm{IC}}/S_{\mathrm{rho}}$ per prime and seed, run-20261018T1548Z.

p	seed	relations	solved, correct	S before	S after	S/S_{rho} before	S/S_{rho} after
53	1	31	yes, yes	4734	936.9	4297	850
53	2	33	yes, yes	2121	425.2	1843	370
101	1	53	yes, yes	1455	287.5	981	194
101	2	50	yes, yes	2370	466.4	1897	373

Both arms see the same residual streams and find the same relations; every logarithm is correct. S_{rho} is the mean of 4 runs (1.10 ± 0.20, 1.15 ± 0.69, 1.48 ± 0.52, 1.25 ± 0.58). Wall time of the whole dlp invocation: 912 s before, 370 s after, including the identical rho runs.

Records

`raw/{ccov,dlp}_{main,new}.{json,log}`, `raw/walls.txt`, `raw/versions.txt`; `../../summary.json` is the frozen summary the leaderboard reads; graphs from `../../plot.py`, which reads `raw/` only.